The Induced Leak

Electromagnetic Side-Channel Attack Through Inductive Coupling

Theoretical, Mathematical, and Physical Foundations

A Research-Level Comprehensive Treatment

Institution: ENSTA Algiers

Department: Computer Science Fundamentals (MI2)

Course Director: Dr. Cheggou Rabéa

Prepared by: Djaber Alae Dine

Academic Year: 2025–2026

April 1, 2026

Contents

1 Introduction **3**
1.1 Motivation and Problem Statement . . . 3
1.2 Scope of This Document . . . 4

2 Maxwell's Equations: Complete Electromagnetic Foundation **4**
2.1 Differential Form in SI Units . . . 4
2.2 Integral Form and Physical Interpretation . . . 5
2.3 Simplification for the Attack Scenario . . . 5
2.4 The Magnetic Vector Potential . . . 6
2.5 Derivation of the Electromagnetic Wave Equation . . . 7

3 Faraday's Law of Electromagnetic Induction: The Attack Mechanism **7**
3.1 General Statement of Faraday's Law . . . 7
3.2 Magnetic Flux Through the Receiving Loop . . . 8
3.3 The Induced Electromotive Force . . . 8
3.4 Faraday's Law for a Finite-Length Cable Segment . . . 9

4 Mutual Inductance: Rigorous Treatment **9**
4.1 Definition and Fundamental Properties . . . 9
4.2 Self-Inductance of a Straight Wire . . . 10
4.3 Parallel Wire Pair: Grover's Formula . . . 10
4.4 Coupling Coefficient . . . 10
4.5 Frequency-Dependent Mutual Inductance . . . 11

5 Digital Signal Model and Electromagnetic Signature **11**
5.1 Binary Signal Representation . . . 11
5.2 The Induced Attack Signal . . . 12
5.3 Realistic Pulse Shaping: Rise Time Effects . . . 12
5.4 Signal Reconstruction Algorithm . . . 13
5.5 Adaptive Threshold for Noise Resilience . . . 13

6 Crosstalk Theory: Near-End and Far-End Coupling **14**
6.1 The Telegrapher's Equations . . . 14
6.2 Coupled Transmission Line Model . . . 15
6.3 Near-End Crosstalk (NEXT) and Far-End Crosstalk (FEXT) . . . 15
6.4 Crosstalk in Twisted Pair Cables . . . 15

7 Fourier Analysis of Electromagnetic Signatures **16**
7.1 Fourier Transform of the Induced Signal . . . 16
7.2 Spectral Content of NRZ Data . . . 16
7.3 Spectral Analysis for Various Line Codes . . . 17
7.4 Bandwidth Considerations for the Attack . . . 17
7.5 Short-Time Fourier Transform (STFT) . . . 17

8 Noise Models and Signal-to-Noise Ratio Analysis **17**
8.1 Sources of Electromagnetic Noise . . . 18
8.1.1 Thermal (Johnson–Nyquist) Noise . . . 18
8.1.2 Shot Noise . . . 18
8.1.3 Electromagnetic Interference (EMI) . . . 18
8.1.4 Quantization Noise . . . 18
8.2 Total Noise Model . . . 18
8.3 Signal-to-Noise Ratio . . . 19
8.4 Bit Error Rate Analysis . . . 19
8.5 Distance-Dependent SNR . . . 19

9 Electromagnetic Shielding Theory **20**
9.1 Shielding Effectiveness: Definition . . . 20
9.2 Shielding Mechanisms . . . 20
9.3 Shielding Effectiveness of Common Materials . . . 20
9.4 Skin Depth Analysis . . . 21
9.5 Shielding Effectiveness at Cable Interfaces . . . 21

10 Independent Component Analysis for Multi-Cable Separation **21**
10.1 The Linear Mixing Model . . . 21
10.2 Statistical Assumptions . . . 22
10.3 Preprocessing: Whitening . . . 22
10.4 FastICA Algorithm . . . 22

11 Transmission Line Theory for Cable Radiation **23**
11.1 Current Mode Analysis . . . 23
11.1.1 Differential Mode (DM) . . . 23
11.1.2 Common Mode (CM) . . . 23
11.2 Antenna Model for Cable Emissions . . . 23

12 Biot–Savart Law and Near-Field Analysis **24**
12.1 Biot–Savart Law . . . 24
12.2 Magnetic Field of a Finite Straight Wire . . . 24
12.3 Near-Field vs. Far-Field Transitions . . . 24

13 Poynting Vector and Energy Flow **25**
13.1 The Poynting Theorem . . . 25
13.2 Time-Averaged Power Flow . . . 25

14 Cryptographic Implications of Physical Leakage **25**
14.1 Information-Theoretic Framework . . . 26
14.2 Side-Channel Attacks on Cryptographic Implementations . . . 26
14.2.1 Simple Power Analysis (SPA) . . . 26
14.2.2 Differential Power Analysis (DPA) . . . 26

15 Conclusion and Open Research Directions **27**
15.1 Summary of Key Theoretical Results . . . 27
15.2 Open Research Problems . . . 27

1 Introduction

The security of digital communication systems has traditionally been analyzed through the lens of cryptographic protocols, network architecture, and software vulnerabilities. However, a fundamentally different class of threats operates at the physical layer, where the electromagnetic behavior of electronic circuits inadvertently leaks information to an external observer. This research domain, known as **electromagnetic side-channel attacks (EM-SCA)**, exploits the principle that any time-varying electric current generates a propagating electromagnetic field, which, under suitable conditions, can be intercepted and decoded without any physical modification to the target system.

The present work, titled *The Induced Leak*, focuses on a specific yet critically important manifestation of this vulnerability: the **inductive coupling** between parallel cables carrying digital data. When current flows through a conductor—such as an Ethernet or USB cable—it creates a magnetic field described by the Biot–Savart law and governed by Maxwell's equations. A second conductor placed in proximity to this field experiences an electromotive force (EMF) by virtue of Faraday's law of induction. If the original current encodes binary data (as is the case for all digital communication protocols), then the induced EMF carries a distorted but recoverable copy of that data. This constitutes a completely passive, non-invasive attack vector that is extraordinarily difficult to detect and counter.

1.1 Motivation and Problem Statement

The central research question addressed in this document can be formulated as follows:

> **Research Question:** Given two parallel conductors in free space (or a bounded medium), under what physical and geometric conditions can the electromagnetic field radiated by time-varying digital currents in one conductor induce a measurable and decodable signal in the other, thereby enabling reconstruction of the original binary data stream?

This question sits at the intersection of three major scientific disciplines:

(i) **Classical Electrodynamics**: The complete description of electromagnetic field generation, propagation, and interaction with matter, as formalized by Maxwell's equations.

(ii) **Circuit Theory and Signal Processing**: The modeling of digital signals, their spectral content, noise characteristics, and reconstruction algorithms.

(iii) **Information Security**: The classification, analysis, and mitigation of physical-layer attack vectors against cryptographic and communication systems.

The importance of this research is underscored by several contemporary developments. First, the proliferation of Internet of Things (IoT) devices has dramatically increased the number of unshielded or poorly shielded data cables deployed in sensitive environments. Second, the advent of high-speed communication standards (10 Gbps Ethernet, USB 4.0, HDMI 2.1) results in stronger electromagnetic emissions at higher frequencies, which are more easily intercepted. Third, the development of quantum-resistant cryptographic algorithms, while crucial for long-term security, provides no protection whatsoever against

physical-layer key extraction—a point frequently overlooked in security assessments. Finally, the increasing density of electromagnetic radiation in urban environments due to 5G/6G infrastructure creates both opportunities for signal masking and challenges for reliable interception.

1.2 Scope of This Document

This document provides a comprehensive, research-level treatment of the theoretical, mathematical, and physical foundations required to understand, model, and analyze electromagnetic side-channel attacks through inductive coupling. Unlike engineering-oriented treatments that focus on measurement techniques and practical countermeasures, we adopt a rigorously theoretical approach, deriving every result from first principles.

The document is organized as follows. Section 2 establishes the complete electromagnetic foundation, beginning with Maxwell's equations in their full differential and integral forms and culminating in the derivation of the wave equation. Section 3 provides an exhaustive treatment of Faraday's law and its application to the attack mechanism. Section 4 develops the theory of mutual inductance, including the Neumann formula, Grover's approximations, and cable-specific configurations. Section 5 addresses the mathematical modeling of digital signals and their electromagnetic signatures. Section 6 extends the analysis to crosstalk theory, including near-end and far-end coupling. Section 7 provides the frequency-domain perspective through Fourier analysis. Section 8 develops the noise and interference models essential for evaluating attack feasibility. Section 9 presents the theory of electromagnetic shielding. Finally, Section 15 summarizes the theoretical framework and identifies open research directions.

2 Maxwell's Equations: Complete Electromagnetic Foundation

The entire edifice of classical electrodynamics rests upon four coupled partial differential equations, collectively known as **Maxwell's equations**. These equations describe the generation, propagation, and interaction of electric and magnetic fields with charges and currents. In the context of electromagnetic side-channel attacks, Maxwell's equations provide the fundamental link between digital data currents and the radiated fields that carry information to an external observer.

2.1 Differential Form in SI Units

In the International System of Units (SI), Maxwell's equations in their general form are:

$$\boldsymbol{\nabla} \cdot \boldsymbol{D} = \rho_f \qquad \text{(Gauss's Law)}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0 \qquad \text{(Gauss's Law for Magnetism)}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \qquad \text{(Faraday's Law)}$$

$$\boldsymbol{\nabla} \times \boldsymbol{H} = \boldsymbol{J}_f + \frac{\partial \boldsymbol{D}}{\partial t} \qquad \text{(Ampère–Maxwell Law)}$$

where the fields are defined as follows:

- $\boldsymbol{E}$: Electric field intensity ($\mathrm{V\,m^{-1}}$)
- $\boldsymbol{D}$: Electric displacement field ($\mathrm{C\,m^{-2}}$)
- $\boldsymbol{B}$: Magnetic flux density (T)
- $\boldsymbol{H}$: Magnetic field intensity ($\mathrm{A\,m^{-1}}$)
- $\boldsymbol{J}_f$: Free current density ($\mathrm{A\,m^{-2}}$)
- ρ_f: Free charge density ($\mathrm{C\,m^{-3}}$)

The constitutive relations in linear, isotropic media are:

$$\boldsymbol{D} = \varepsilon \boldsymbol{E} = \varepsilon_0 \varepsilon_r \boldsymbol{E} \tag{1}$$

$$\boldsymbol{B} = \mu \boldsymbol{H} = \mu_0 \mu_r \boldsymbol{H} \tag{2}$$

$$\boldsymbol{J}_f = \sigma \boldsymbol{E} \tag{3}$$

where $\varepsilon_0 = 8.854 \times 10^{-12}\,\mathrm{F/m}$ is the vacuum permittivity, $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H/m}$ is the vacuum permeability, ε_r is the relative permittivity, μ_r is the relative permeability, and σ is the electrical conductivity.

2.2 Integral Form and Physical Interpretation

The integral forms of Maxwell's equations reveal the physical mechanisms at play:

$$\oint_S \boldsymbol{D} \cdot \mathrm{d}\boldsymbol{A} = \int_V \rho_f \mathrm{d}V \tag{Gauss}$$

$$\oint_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{A} = 0 \tag{No monopoles}$$

$$\oint_C \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{\ell} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{A} \tag{Faraday}$$

$$\oint_C \boldsymbol{H} \cdot \mathrm{d}\boldsymbol{\ell} = \int_S \boldsymbol{J}_f \cdot \mathrm{d}\boldsymbol{A} + \frac{\mathrm{d}}{\mathrm{d}t} \int_S \boldsymbol{D} \cdot \mathrm{d}\boldsymbol{A} \tag{Ampère}$$

Remark 2.1. *The physical interpretation of Faraday's law (equation 15) is the cornerstone of the electromagnetic side-channel attack. It states that a time-varying magnetic flux through any closed loop induces an electromotive force around that loop, regardless of whether a physical conductor exists along the loop path. In the context of the attack, the "closed loop" is formed by the attacker's receiving antenna or sensing cable, and the time-varying flux is generated by the digital data current flowing through the victim cable.*

2.3 Simplification for the Attack Scenario

In the specific scenario relevant to electromagnetic side-channel attacks on cables, we make the following physically justified simplifications:

S1: No free charges: In the region between the victim and attacker cables, $\rho_f = 0$, since all charges reside within the conductors themselves.

S2: Linear, isotropic medium: The space between cables is assumed to be air (or vacuum), so $\varepsilon_r \approx 1$, $\mu_r \approx 1$, and $\sigma \approx 0$.

S3: Quasi-static approximation: For the frequencies of interest (up to several GHz for modern data cables), the displacement current term $\varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}$ is negligible compared to the conduction current $\boldsymbol{J}_f$ inside the conductors. However, in the space between conductors where $\boldsymbol{J}_f = 0$, the displacement current is the sole contributor to the curl of $\boldsymbol{H}$.

Under these conditions, Maxwell's equations simplify to:

$$\begin{aligned}
\boldsymbol{\nabla} \cdot \boldsymbol{E} &= 0 && \text{(in free space)} \\
\boldsymbol{\nabla} \cdot \boldsymbol{B} &= 0 && (4) \\
\boldsymbol{\nabla} \times \boldsymbol{E} &= -\frac{\partial \boldsymbol{B}}{\partial t} && \text{(Faraday's Law)} \\
\boldsymbol{\nabla} \times \boldsymbol{B} &= \mu_0 \boldsymbol{J}_f + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} && \text{(Ampère–Maxwell)}
\end{aligned}$$

2.4 The Magnetic Vector Potential

To solve Maxwell's equations for practical cable geometries, we introduce the **magnetic vector potential** $\boldsymbol{A}$, defined such that:

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A} \tag{5}$$

This definition automatically satisfies Gauss's law for magnetism ($\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$) since $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) \equiv 0$. Substituting into Faraday's law:

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \boldsymbol{A}) = -\boldsymbol{\nabla} \times \left(\frac{\partial \boldsymbol{A}}{\partial t} \right) \tag{6}$$

which implies:

$$\boldsymbol{\nabla} \times \left(\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t} \right) = \boldsymbol{0} \tag{7}$$

Since the curl of any gradient is identically zero, we can write:

$$\boldsymbol{E} = -\nabla \phi - \frac{\partial \boldsymbol{A}}{\partial t} \tag{8}$$

where ϕ is the scalar electric potential. In the Coulomb gauge ($\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0$) for static or quasi-static situations, the vector potential in free space due to a current distribution $\boldsymbol{J}(\boldsymbol{r}', t)$ is given by:

$$\boldsymbol{A}(\boldsymbol{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{r}', t_{\text{ret}})}{|\boldsymbol{r} - \boldsymbol{r}'|} \mathrm{d}V' \tag{9}$$

where $t_{\text{ret}} = t - |\boldsymbol{r} - \boldsymbol{r}'|/c$ is the retarded time and $c = 1/\sqrt{\mu_0 \varepsilon_0} \approx 3 \times 10^8\,\mathrm{m/s}$ is the speed of light. For the quasi-static approximation relevant to our attack scenario ($|\boldsymbol{r} - \boldsymbol{r}'| \ll c/f$, where f is the signal frequency), we can neglect the retardation effect:

$$\boldsymbol{A}(\boldsymbol{r}) \approx \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} \mathrm{d}V' \tag{10}$$

2.5 Derivation of the Electromagnetic Wave Equation

Taking the curl of Faraday's law (equation 20):

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}) = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \boldsymbol{B}) = -\mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t} - \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2} \tag{11}$$

Using the vector identity $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}) = \nabla(\boldsymbol{\nabla} \cdot \boldsymbol{E}) - \nabla^2 \boldsymbol{E}$ and $\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0$ in free space:

$$\nabla^2 \boldsymbol{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2} = \mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t} \tag{12}$$

Similarly, for the magnetic field:

$$\nabla^2 \boldsymbol{B} - \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{B}}{\partial t^2} = -\mu_0 \boldsymbol{\nabla} \times \boldsymbol{J}_f \tag{13}$$

In source-free regions ($\boldsymbol{J}_f = \boldsymbol{0}$), these reduce to the homogeneous wave equations:

$$\nabla^2 \boldsymbol{E} - \frac{1}{c^2} \frac{\partial^2 \boldsymbol{E}}{\partial t^2} = \boldsymbol{0} \tag{14}$$

$$\nabla^2 \boldsymbol{B} - \frac{1}{c^2} \frac{\partial^2 \boldsymbol{B}}{\partial t^2} = \boldsymbol{0} \tag{15}$$

Theorem 2.1 (Wave Propagation Speed)**.** *The wave equations* (**??**) *and* (**??**) *admit solutions of the form* $\boldsymbol{E}(\boldsymbol{r}, t) = \boldsymbol{E}_0 f(\hat{\boldsymbol{n}} \cdot \boldsymbol{r} - ct)$*, which propagate in the direction* $\hat{\boldsymbol{n}}$ *at speed* $c = 1/\sqrt{\mu_0 \varepsilon_0} \approx 2.998 \times 10^8$ *m/s. This propagation speed is a fundamental constant of nature and connects the electromagnetic attack mechanism to the finite speed of information transfer.*

3 Faraday's Law of Electromagnetic Induction: The Attack Mechanism

Faraday's law of electromagnetic induction is the single most important physical principle underlying the electromagnetic side-channel attack. It establishes the quantitative relationship between a time-varying magnetic field and the electromotive force it induces in a circuit. This section provides an exhaustive treatment of Faraday's law, starting from its general statement and progressively specializing to the cable geometry relevant to the attack.

3.1 General Statement of Faraday's Law

Theorem 3.1 (Faraday's Law of Induction)**.** *The electromotive force (EMF) induced in any closed circuit* C *bounding a surface* S *is equal to the negative rate of change of the magnetic flux through* S*:*

$$\mathcal{E} = \oint_C \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{\ell} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t} \tag{16}$$

where the magnetic flux Φ_B *is defined as:*

$$\Phi_B = \int_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{A} = \int_S \boldsymbol{B} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}A \tag{17}$$

Remark 3.1 (The Negative Sign and Lenz's Law)**.** *The negative sign in Faraday's law is not merely a convention; it embodies* ***Lenz's law****, which states that the direction of the induced EMF is such that it opposes the change in magnetic flux that produced it. In the attack context, this means that the induced signal in the attacker's cable always has a polarity opposite to the rate of change of the victim's signal—a crucial detail for correct bit reconstruction.*

3.2 Magnetic Flux Through the Receiving Loop

Consider two parallel, infinitely long, straight cylindrical conductors: the **victim cable** (carrying current $I_v(t)$) and the **receiver cable** (modeled as a rectangular loop of width w and height h). The victim cable lies along the z-axis, and the center of the receiver loop is at distance d from the victim cable.

The magnetic field at distance r from an infinite straight wire carrying current I is, by Ampère's law:

$$\boldsymbol{B}(r) = \frac{\mu_0 I}{2\pi r}\,\hat{\boldsymbol{\phi}} \tag{18}$$

For a rectangular receiving loop whose nearest edge is at distance $d-w/2$ and farthest edge at distance $d+w/2$ from the victim cable (assuming the loop is oriented perpendicular to the radial direction), the magnetic flux through the loop is:

$$\Phi_B = \int_S \boldsymbol{B}\cdot \mathrm{d}\boldsymbol{A} = \frac{\mu_0 I_v(t) h}{2\pi}\int_{d-w/2}^{d+w/2}\frac{1}{r}\,\mathrm{d}r = \frac{\mu_0 I_v(t) h}{2\pi}\ln\left(\frac{d+w/2}{d-w/2}\right) \tag{19}$$

Definition 3.1 (Effective Coupling Width)**.** *We define the* ***effective coupling width*** w_{eff} *such that:*

$$w_{\mathit{eff}} = 2d\sinh^{-1}\left(\frac{w}{2d}\right) \tag{20}$$

which captures the geometric dependence of the magnetic flux on the receiver dimensions relative to the separation distance.

3.3 The Induced Electromotive Force

Applying Faraday's law to (19), the induced EMF is:

$$\mathcal{E}(t) = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t} = -\frac{\mu_0 h}{2\pi}\ln\left(\frac{d+w/2}{d-w/2}\right)\frac{\mathrm{d}I_v(t)}{\mathrm{d}t} \tag{21}$$

Proposition 3.2 (Linearity of Induction)**.** *The induced EMF is* ***proportional to the time derivative*** *of the victim current, not to the current itself. This has profound implications for the attack: constant (DC) currents produce zero EMF, while rapidly switching (high-frequency) currents produce the strongest signals. Consequently, the attack is inherently sensitive to the* ***transition edges*** *of digital signals rather than their steady-state levels.*

For a rectangular pulse train representing non-return-to-zero (NRZ) digital data:

$$I_v(t) = \sum_{n=-\infty}^{\infty} a_n \cdot \mathrm{rect}\left(\frac{t-nT_b}{T_b}\right) \tag{22}$$

where $a_n \in \{I_L, I_H\}$ represents the binary levels and T_b is the bit period, the time derivative $\mathrm{d}I_v/\mathrm{d}t$ consists of delta-like impulses at each bit transition. This explains why the induced signal consists of positive and negative pulses corresponding to rising and falling edges, respectively.

3.4 Faraday's Law for a Finite-Length Cable Segment

In practice, cables have finite length L. The mutual inductance approach provides a more general framework. For two parallel cylindrical conductors of length L separated by distance d, with the receiver modeled as a circular loop of radius a coaxial with a section of the victim cable:

$$\mathcal{E} = -M\frac{\mathrm{d}I_v}{\mathrm{d}t} \tag{23}$$

where M is the mutual inductance, given by the Neumann formula:

$$M = \frac{\mu_0}{4\pi}\oint_{C_1}\oint_{C_2}\frac{\mathrm{d}\boldsymbol{\ell}_1 \cdot \mathrm{d}\boldsymbol{\ell}_2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} \tag{24}$$

For the specific geometry of two parallel wires of length L separated by distance d:

$$M = \frac{\mu_0 L}{2\pi}\left[\ln\left(\frac{L}{d} + \sqrt{1 + \frac{L^2}{d^2}}\right) - \sqrt{1 + \frac{d^2}{L^2}} + \frac{d}{L}\right] \tag{25}$$

Corollary 3.3 (Long-Cable Approximation)**.** *For $L \gg d$ (the typical case for cable runs), (25) simplifies to:*

$$M \approx \frac{\mu_0 L}{2\pi}\ln\left(\frac{2L}{d}\right) \tag{26}$$

This logarithmic dependence on L/d is a key result: the mutual inductance grows only slowly with cable length and decreases only slowly with separation distance, which is why electromagnetic side-channel attacks remain feasible at non-trivial distances.

4 Mutual Inductance: Rigorous Treatment

Mutual inductance quantifies the magnetic coupling between two circuits. A thorough understanding of mutual inductance is essential for predicting the strength of the induced signal and, consequently, the feasibility of the electromagnetic side-channel attack.

4.1 Definition and Fundamental Properties

Definition 4.1 (Mutual Inductance)**.** *The mutual inductance M_{12} between circuit 1 (the source/victim) and circuit 2 (the receiver/attacker) is defined as:*

$$M_{12} = \frac{\Phi_{12}}{I_1} \tag{27}$$

where Φ_{12} is the magnetic flux through circuit 2 due to the current I_1 in circuit 1.

Theorem 4.1 (Reciprocity of Mutual Inductance)**.**

$$M_{12} = M_{21} \equiv M \tag{28}$$

This symmetry is a consequence of the Neumann formula (24) and implies that the coupling from victim to attacker is identical to the coupling from attacker to victim.

Theorem 4.2 (Neumann Formula)**.** *For two arbitrary filamentary circuits C_1 and C_2:*

$$M = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{\mathrm{d}\boldsymbol{\ell}_1 \cdot \mathrm{d}\boldsymbol{\ell}_2}{R_{12}} \tag{29}$$

where $R_{12} = |\boldsymbol{r}_1 - \boldsymbol{r}_2|$ is the distance between the differential elements $\mathrm{d}\boldsymbol{\ell}_1$ and $\mathrm{d}\boldsymbol{\ell}_2$.

4.2 Self-Inductance of a Straight Wire

Before computing mutual inductance, we establish the self-inductance of the victim cable, which determines its own electromagnetic emission characteristics.

For a straight cylindrical conductor of radius a, length L, and carrying uniform current density, the internal and external magnetic flux contributions lead to:

$$L_{\text{self}} = \frac{\mu_0 L}{8\pi} + \frac{\mu_0 L}{2\pi} \ln\left(\frac{2L}{a} - 1\right) \approx \frac{\mu_0 L}{2\pi}\left(\ln\frac{2L}{a} + \frac{1}{4}\right) \tag{30}$$

for $L \gg a$.

4.3 Parallel Wire Pair: Grover's Formula

For the canonical geometry of two parallel wires of radii a_1, a_2, separated by center-to-center distance d (with $d \gg a_1, a_2$), running in parallel for length L:

$$M = \frac{\mu_0}{2\pi}\left[L \cdot \ln\left(\frac{L + \sqrt{L^2 + d^2}}{d}\right) - \sqrt{L^2 + d^2} + d\right] \tag{31}$$

Table 1: Mutual inductance for various cable configurations ($\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H/m}$).

Configuration	L (m)	d (cm)	M (μH)	$\mathcal{E}$ at $1\,\mathrm{Gbit\,s^{-1}}$
Short patch cable	1	1	0.10	0.63 mV
Desktop Ethernet	2	5	0.04	0.25 mV
Server room run	10	10	0.17	1.07 mV
Under-floor bundle	20	2	2.00	12.57 mV
Adjacent in conduit	5	0.5	0.62	3.90 mV

4.4 Coupling Coefficient

The **coupling coefficient** k quantifies the fraction of magnetic flux from one circuit that links the other:

$$k = \frac{M}{\sqrt{L_1 L_2}}, \qquad 0 \le k \le 1 \tag{32}$$

For loosely coupled cables (the typical scenario in side-channel attacks), $k \ll 1$, typically ranging from 10^{-4} to 10^{-2}. Despite this weak coupling, the induced signal can be recovered through signal amplification and processing techniques described in subsequent sections.

4.5 Frequency-Dependent Mutual Inductance

At higher frequencies, the proximity effect and skin effect modify the current distribution within the conductors, leading to a frequency-dependent mutual inductance:

$$M(f) = M_0 \cdot F_{\text{skin}}(f) \cdot F_{\text{prox}}(f) \tag{33}$$

where M_0 is the DC mutual inductance, $F_{\text{skin}}(f)$ is the skin effect correction factor, and $F_{\text{prox}}(f)$ is the proximity effect correction factor. The skin depth is:

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} = \sqrt{\frac{1}{\pi f \mu\sigma}} \tag{34}$$

For copper ($\sigma = 5.96 \times 10^7\,\text{S/m}$) at $f = 1\,\text{GHz}$:

$$\delta = \sqrt{\frac{1}{\pi \times 10^9 \times 4\pi \times 10^{-7} \times 5.96 \times 10^7}} \approx 2.06\,\mu\text{m} \tag{35}$$

5 Digital Signal Model and Electromagnetic Signature

The information content of the electromagnetic side-channel attack resides in the temporal structure of the current flowing through the victim cable. This section develops a rigorous mathematical model for digital signals and their electromagnetic signatures.

5.1 Binary Signal Representation

A binary data stream is represented as a sequence of bits $\{b_n\}_{n=0}^{N-1}$ where $b_n \in \{0, 1\}$. The corresponding electrical signal using NRZ (Non-Return-to-Zero) line coding is:

$$I(t) = \sum_{n=0}^{N-1} a_n \cdot p(t - nT_b) \tag{36}$$

where:

- $a_n \in \{I_L, I_H\}$ is the current level corresponding to bit b_n (e.g., $I_L = -5\,\text{mA}$, $I_H = +5\,\text{mA}$)
- $T_b = 1/R_b$ is the bit period, where R_b is the data rate in bits per second
- $p(t)$ is the pulse shape function, typically a rectangular pulse:

$$p(t) = \begin{cases} 1 & \text{if } 0 \le t < T_b \\ 0 & \text{otherwise} \end{cases} \tag{37}$$

5.2 The Induced Attack Signal

The induced EMF at the attacker's receiver is obtained by applying Faraday's law to the victim's current signal:

$$\mathcal{E}(t) = -M\frac{\mathrm{d}I(t)}{\mathrm{d}t} = -M\sum_{n=0}^{N-1} a_n \cdot p'(t - nT_b) \tag{38}$$

where $p'(t) = \mathrm{d}p(t)/\mathrm{d}t$ is the derivative of the pulse shape. For the rectangular pulse, the derivative consists of two delta functions at the rising and falling edges:

$$p'(t) = \delta(t) - \delta(t - T_b) \tag{39}$$

Therefore, the induced signal becomes:

$$\mathcal{E}(t) = -M\sum_{n=0}^{N-1}(a_n - a_{n-1}) \cdot \delta(t - nT_b) \tag{40}$$

where $a_{-1} = 0$ by convention (assuming the line is idle before transmission).

Proposition 5.1 (Transition Encoding)**.** *Equation (40) reveals that the induced EMF encodes the* ***differences*** *between consecutive bits, not the absolute bit values. The quantity $\Delta_n = a_n - a_{n-1}$ has four possible values:*

b_{n-1}	b_n	Δ_n	Physical Meaning
0	0	0	No transition (no EMF pulse)
0	1	$I_H - I_L$	Rising edge (positive EMF pulse)
1	0	$I_L - I_H$	Falling edge (negative EMF pulse)
1	1	0	No transition (no EMF pulse)

This differential encoding is fundamental to the attack: the attacker observes ***transitions****, not levels, and must integrate the transitions to reconstruct the original data stream.*

5.3 Realistic Pulse Shaping: Rise Time Effects

In practice, digital signals do not have zero rise time. A more realistic model uses a trapezoidal pulse with rise time τ_r and fall time τ_f:

$$p(t) = \begin{cases} \frac{t}{\tau_r} & 0 \le t < \tau_r \\ 1 & \tau_r \le t < T_b - \tau_f \\ \frac{T_b - t}{\tau_f} & T_b - \tau_f \le t < T_b \\ 0 & \text{otherwise} \end{cases} \tag{41}$$

The derivative of this trapezoidal pulse gives the induced signal shape:

$$p'(t) = \begin{cases} \frac{1}{\tau_r} & 0 \le t < \tau_r \\ 0 & \tau_r \le t < T_b - \tau_f \\ -\frac{1}{\tau_f} & T_b - \tau_f \le t < T_b \\ 0 & \text{otherwise} \end{cases} \tag{42}$$

Remark 5.1 (Bandwidth Implications)**.** *The rise time τ_r determines the signal bandwidth. For a trapezoidal pulse, the 3 dB bandwidth is approximately:*

$$BW \approx \frac{0.35}{\tau_r} \tag{43}$$

Modern high-speed interfaces achieve rise times as short as 30 ps*, corresponding to bandwidths exceeding* 10 GHz*. This is precisely why high-speed data cables are more vulnerable: their electromagnetic signatures extend to higher frequencies where the induced EMF is stronger.*

5.4 Signal Reconstruction Algorithm

The reconstruction of the original bit sequence from the induced EMF signal requires solving an inverse problem. We formalize this as follows:

Definition 5.1 (Bit Reconstruction Problem)**.** *Given a sampled induced signal $\{\mathcal{E}[k]\}_{k=0}^{K-1}$ with known bit period T_b and mutual inductance M, find the bit sequence $\{b_n\}_{n=0}^{N-1}$ that maximizes the likelihood of the observation under a Gaussian noise model.*

A practical reconstruction algorithm operates as follows:

Input : Sampled EMF signal $\{\mathcal{E}[k]\}$, sampling rate f_s, bit period T_b, threshold τ
Output: Reconstructed bit sequence $\{\hat{b}_n\}$

```
N_b ← ⌊f_s · T_b⌋
samples_per_bit ← N_b
n ← 0
for k ← 0 to K − 1 step N_b do
    max_val ← max(E[k : k + N_b])
    min_val ← min(E[k : k + N_b])
    if max_val > +τ then
        b̂_n ← 1
    end
    else if min_val < −τ then
        b̂_n ← 0
    end
    else
        b̂_n ← b̂_{n−1}                    // No transition detected
    end
    n ← n + 1
end
return {b̂_n}
```

Algorithm 1: Basic Edge-Detection Bit Reconstruction

5.5 Adaptive Threshold for Noise Resilience

The fixed threshold in Algorithm 1 fails under varying noise conditions. An adaptive threshold based on local statistics provides robustness:

$$\tau(t) = \mu_{\text{local}}(t) + k \cdot \sigma_{\text{local}}(t) \tag{44}$$

where μ_{local} and σ_{local} are the local mean and standard deviation computed over a sliding window of duration W, and k is an adaptation coefficient (typically $k \in [2, 4]$ for a 95–99.9% confidence level under Gaussian noise assumptions).

The local statistics are computed as:

$$\mu_{\text{local}}(t) = \frac{1}{W} \int_{t-W/2}^{t+W/2} \mathcal{E}(\tau)\, \mathrm{d}\tau \tag{45}$$

$$\sigma^2_{\text{local}}(t) = \frac{1}{W} \int_{t-W/2}^{t+W/2} \left[\mathcal{E}(\tau) - \mu_{\text{local}}(t)\right]^2 \mathrm{d}\tau \tag{46}$$

6 Crosstalk Theory: Near-End and Far-End Coupling

Crosstalk is the unintended electromagnetic coupling between adjacent signal paths. In the context of our attack, the crosstalk from the victim cable to the attacker's receiver is precisely the attack channel. This section develops the transmission line theory necessary to model this coupling rigorously.

6.1 The Telegrapher's Equations

A uniform transmission line is characterized by four distributed parameters per unit length:

- R': Series resistance ($\Omega\,\mathrm{m}^{-1}$)
- L': Series inductance ($\mathrm{H\,m}^{-1}$)
- G': Shunt conductance ($\mathrm{S\,m}^{-1}$)
- C': Shunt capacitance ($\mathrm{F\,m}^{-1}$)

The voltage and current on the line satisfy the telegrapher's equations:

$$\frac{\partial V(z,t)}{\partial z} = -R' I(z,t) - L' \frac{\partial I(z,t)}{\partial t} \qquad \text{(Voltage equation)}$$

$$\frac{\partial I(z,t)}{\partial z} = -G' V(z,t) - C' \frac{\partial V(z,t)}{\partial t} \qquad \text{(Current equation)}$$

For lossless lines ($R' = 0$, $G' = 0$):

$$\frac{\partial V}{\partial z} = -L' \frac{\partial I}{\partial t} \tag{47}$$

$$\frac{\partial I}{\partial z} = -C' \frac{\partial V}{\partial t} \tag{48}$$

Combining these yields the wave equation for both V and I:

$$\frac{\partial^2 V}{\partial z^2} = L'C' \frac{\partial^2 V}{\partial t^2}, \qquad \frac{\partial^2 I}{\partial z^2} = L'C' \frac{\partial^2 I}{\partial t^2} \tag{49}$$

with the propagation velocity:

$$v_p = \frac{1}{\sqrt{L'C'}}, \qquad \text{and the characteristic impedance: } Z_0 = \sqrt{\frac{L'}{C'}} \tag{50}$$

6.2 Coupled Transmission Line Model

For two parallel transmission lines (victim and receiver), the coupling introduces additional terms:

$$\frac{\partial V_1}{\partial z} = -L'_{11}\frac{\partial I_1}{\partial t} - L'_{12}\frac{\partial I_2}{\partial t} \tag{51}$$

$$\frac{\partial V_2}{\partial z} = -L'_{21}\frac{\partial I_1}{\partial t} - L'_{22}\frac{\partial I_2}{\partial t} \tag{52}$$

$$\frac{\partial I_1}{\partial z} = -C'_{11}\frac{\partial V_1}{\partial t} - C'_{12}\frac{\partial V_2}{\partial t} \tag{53}$$

$$\frac{\partial I_2}{\partial z} = -C'_{21}\frac{\partial V_1}{\partial t} - C'_{22}\frac{\partial V_2}{\partial t} \tag{54}$$

where $L'_{12} = L'_{21} = M'$ is the mutual inductance per unit length and $C'_{12} = C'_{21}$ is the mutual capacitance per unit length.

6.3 Near-End Crosstalk (NEXT) and Far-End Crosstalk (FEXT)

In the frequency domain, for a coupled line segment of length ℓ with the victim driven at $z = 0$ and terminated at both ends:

Near-End Crosstalk (NEXT) — measured at $z = 0$ on the receiver:

$$\mathrm{NEXT}(f) = \frac{j\omega M'}{2}\frac{1 - e^{-2\gamma\ell}}{\gamma(1 - \Gamma_S\Gamma_L e^{-2\gamma\ell})} + \frac{j\omega C' Z'_0}{2}\frac{1 - e^{-2\gamma\ell}}{\gamma(1 - \Gamma_S\Gamma_L e^{-2\gamma\ell})} \tag{55}$$

Far-End Crosstalk (FEXT) — measured at $z = \ell$ on the receiver:

$$\mathrm{FEXT}(f) = j\omega\ell e^{-\gamma\ell}\left(\frac{M'}{2Z_0} - \frac{C' Z'_0}{2}\right) \tag{56}$$

where $\gamma = \alpha + j\beta$ is the propagation constant, Γ_S and Γ_L are the source and load reflection coefficients, and Z'_0 is the characteristic impedance of the receiver line.

Theorem 6.1 (FEXT Scaling)**.** *For electrically short lines ($\beta\ell \ll 1$) and matched terminations:*

$$|FEXT| \propto f \cdot \ell \cdot k_c \tag{57}$$

where k_c is the coupling coefficient. This linear scaling with both frequency and length makes long, high-speed cable runs particularly vulnerable.

6.4 Crosstalk in Twisted Pair Cables

Ethernet cables use twisted pairs to reduce crosstalk. The twisting introduces a spatially alternating coupling polarity that cancels the net induced signal over each twist cycle. For a twist pitch P (length of one complete twist), the cancellation factor is approximately:

$$F_{\text{twist}} \approx \frac{P}{2\pi d} \tag{58}$$

For Category 6 Ethernet cable with $P \approx 1.5\,\text{cm}$ and $d \approx 0.5\,\text{cm}$:

$$F_{\text{twist}} \approx \frac{1.5}{2\pi \times 0.5} \approx 0.48 \tag{59}$$

However, twist cancellation is imperfect. Residual crosstalk arises from:

1. Variation in twist pitch along the cable length
2. Non-uniform separation between pairs
3. Untwisted sections at connectors and patch panels
4. Mechanical deformation during installation

7 Fourier Analysis of Electromagnetic Signatures

The frequency-domain perspective provided by Fourier analysis is indispensable for understanding the spectral content of electromagnetic emissions and designing effective signal processing pipelines for the attack.

7.1 Fourier Transform of the Induced Signal

The Fourier transform of the induced EMF signal (38) is:

$$\mathcal{F}\{\mathcal{E}(t)\} = -j\omega M \cdot I(\omega) \equiv \mathcal{E}(\omega) \tag{60}$$

where $I(\omega) = \mathcal{F}\{I(t)\}$ is the Fourier transform of the victim current. The factor $-j\omega$ confirms that differentiation in the time domain corresponds to multiplication by $j\omega$ in the frequency domain, and the magnitude of the induced signal increases linearly with frequency.

7.2 Spectral Content of NRZ Data

For a random NRZ bit sequence with bit rate $R_b = 1/T_b$, the power spectral density (PSD) is:

$$S_I(f) = A^2 T_b \operatorname{sinc}^2(fT_b) \tag{61}$$

where $A = I_H - I_L$ is the current swing and $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$.

The PSD of the induced EMF is therefore:

$$S_{\mathcal{E}}(f) = \omega^2 M^2 S_I(f) = 4\pi^2 f^2 M^2 A^2 T_b \operatorname{sinc}^2(fT_b) \tag{62}$$

Proposition 7.1 (Spectral Peak)**.** *The function $f^2 \operatorname{sinc}^2(fT_b)$ attains its maximum at approximately $f \approx 0.42/T_b = 0.42R_b$. For a $1\,\mathrm{Gbit\,s^{-1}}$ data stream, the peak spectral component of the induced signal occurs at approximately $420\,\mathrm{MHz}$. This frequency determines the optimal center frequency for the attacker's bandpass filter.*

7.3 Spectral Analysis for Various Line Codes

Table 2: Power spectral density of common line codes. T_b is the bit period, A is the signal amplitude.

Line Code	**Power Spectral Density** $S(f)$
NRZ (Unipolar)	$\frac{A^2 T_b}{4}\left[\mathrm{sinc}^2(fT_b) + \frac{1}{T_b}\delta(f)\right]$
NRZ (Polar)	$A^2 T_b \,\mathrm{sinc}^2(fT_b)$
RZ (50% duty)	$\frac{A^2 T_b}{4}\mathrm{sinc}^2\left(\frac{fT_b}{2}\right)\left[1 + \frac{1}{T_b}\delta\left(f - \frac{1}{T_b}\right)\right]$
Manchester	$A^2 T_b \frac{\sin^4(\pi f T_b/2)}{(\pi f T_b/2)^2}$
4B/5B (MLT-3)	$\frac{5A^2 T_b}{4}\mathrm{sinc}^2\left(\frac{5fT_b}{4}\right)\cos^2\left(\frac{5\pi f T_b}{4}\right)$

7.4 Bandwidth Considerations for the Attack

The attack receiver must capture sufficient bandwidth to resolve individual bit transitions. The Nyquist criterion requires:

$$f_s \geq 2B_{\text{signal}} \tag{63}$$

For NRZ data at rate R_b, the null-to-null bandwidth is $2R_b$, so:

$$f_s \geq 4R_b \tag{64}$$

In practice, a sampling rate of $f_s = 10R_b$ is recommended to provide margin for filter roll-off and to enable accurate edge timing estimation.

7.5 Short-Time Fourier Transform (STFT)

For non-stationary signals (which digital data inherently is), the Short-Time Fourier Transform provides time-frequency localization:

$$\mathrm{STFT}\{\mathcal{E}(t)\}(\tau, f) = \int_{-\infty}^{\infty} \mathcal{E}(t) w(t-\tau) e^{-j2\pi f t}\,\mathrm{d}t \tag{65}$$

where $w(t)$ is a window function (e.g., Hamming, Hanning, or Gaussian). The STFT allows the attacker to identify temporal patterns in the electromagnetic emission, such as packet boundaries, protocol headers, and key exchange sequences.

8 Noise Models and Signal-to-Noise Ratio Analysis

The feasibility of an electromagnetic side-channel attack depends critically on the signal-to-noise ratio (SNR) at the receiver. This section develops comprehensive noise models for the attack scenario and derives the theoretical bounds on attack performance.

8.1 Sources of Electromagnetic Noise

8.1.1 Thermal (Johnson–Nyquist) Noise

Thermal noise is generated by the random thermal motion of charge carriers in any resistive element:

$$V_n^2 = 4k_B T R \Delta f \tag{66}$$

where $k_B = 1.381 \times 10^{-23}\,\mathrm{J/K}$ is Boltzmann's constant, T is the absolute temperature in Kelvin, R is the resistance in ohms, and Δf is the measurement bandwidth.

For a receiver with $R = 50\,\Omega$ at room temperature ($T = 300\,\mathrm{K}$) and bandwidth $\Delta f = 1\,\mathrm{GHz}$:

$$V_n = \sqrt{4 \times 1.381 \times 10^{-23} \times 300 \times 50 \times 10^9} \approx 28.8\,\mu\mathrm{V} \tag{67}$$

8.1.2 Shot Noise

Shot noise arises from the discrete nature of electric charge in active devices:

$$I_n^2 = 2qI_{\mathrm{DC}}\Delta f \tag{68}$$

where $q = 1.602 \times 10^{-19}\,\mathrm{C}$ is the electron charge and I_{DC} is the DC bias current.

8.1.3 Electromagnetic Interference (EMI)

Environmental EMI includes:

- Power line harmonics (50/60 Hz and multiples)
- Radio frequency interference from wireless devices
- Switching noise from power supplies (typically 100 kHz–10 MHz)
- Cosmic background radiation ($\approx 3\,\mathrm{K}$ equivalent)

8.1.4 Quantization Noise

For an analog-to-digital converter (ADC) with n bits and full-scale range V_{FS}:

$$V_{q,\mathrm{rms}} = \frac{V_{\mathrm{FS}}}{2^n\sqrt{12}} \tag{69}$$

For a 12-bit ADC with $V_{\mathrm{FS}} = 1\,\mathrm{V}$:

$$V_{q,\mathrm{rms}} = \frac{1}{2^{12}\sqrt{12}} \approx 70.5\,\mu\mathrm{V} \tag{70}$$

8.2 Total Noise Model

Assuming the noise sources are statistically independent, the total noise power is the sum of individual noise powers:

$$P_{\mathrm{noise}} = P_{\mathrm{thermal}} + P_{\mathrm{shot}} + P_{\mathrm{EMI}} + P_{\mathrm{quantization}} \tag{71}$$

In practice, thermal noise and EMI dominate in most attack scenarios. We model the total noise as additive white Gaussian noise (AWGN):

$$n(t) \sim \mathcal{N}(0, \sigma_n^2), \qquad \sigma_n^2 = N_0 B_w \tag{72}$$

where $N_0/2$ is the two-sided noise power spectral density and B_w is the receiver bandwidth.

8.3 Signal-to-Noise Ratio

The signal-to-noise ratio at the receiver output is:

$$\mathrm{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{M^2\overline{(\mathrm{d}I/\mathrm{d}t)^2}}{N_0 B_w} \tag{73}$$

For NRZ data with bit rate R_b, the mean-square derivative of the current is:

$$\overline{\left(\frac{\mathrm{d}I}{\mathrm{d}t}\right)^2} \approx \frac{2A^2}{3\tau_r^2} \cdot p_{\text{transition}} \tag{74}$$

where $p_{\text{transition}}$ is the transition probability (typically $p_{\text{transition}} \approx 0.5$ for random data).

8.4 Bit Error Rate Analysis

Under the AWGN assumption, the probability of bit error for the edge-detection reconstruction is:

$$P_b = Q\left(\sqrt{\mathrm{SNR}}\right) = \frac{1}{2}\,\mathrm{erfc}\left(\sqrt{\frac{\mathrm{SNR}}{2}}\right) \tag{75}$$

where $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2}\,\mathrm{d}u$ is the Q-function.

Proposition 8.1 (Minimum SNR for Feasible Attack)**.** *For a practically useful attack (target BER $< 10^{-3}$):*

$$SNR > Q^{-1}(10^{-3})^2 \approx (3.09)^2 \approx 9.55 \approx 9.8\,\mathrm{dB} \tag{76}$$

For high-fidelity reconstruction (BER $< 10^{-6}$):

$$SNR > Q^{-1}(10^{-6})^2 \approx (4.75)^2 \approx 22.6 \approx 13.5\,\mathrm{dB} \tag{77}$$

8.5 Distance-Dependent SNR

Since the mutual inductance decreases with distance, the SNR is a function of the victim-receiver separation d. For the parallel wire configuration with $L \gg d$:

$$\mathrm{SNR}(d) = \frac{M(d)^2\overline{(\mathrm{d}I/\mathrm{d}t)^2}}{N_0 B_w} \propto \frac{1}{d^2}\ln^2\left(\frac{2L}{d}\right) \tag{78}$$

Table 3: Estimated SNR as a function of distance for a $1\,\mathrm{Gbit\,s^{-1}}$ Ethernet signal ($L = 10\,\mathrm{m}$, $A = 10\,\mathrm{mA}$).

Distance d (cm)	**SNR (dB)**	**BER (estimated)**
1	42.3	$< 10^{-30}$
2	36.1	$< 10^{-24}$
5	28.4	$< 10^{-14}$
10	22.7	$\approx 10^{-10}$
20	17.0	$\approx 10^{-6}$
50	10.3	$\approx 10^{-3}$
100	4.6	$\approx 2 \times 10^{-2}$
200	-1.1	> 0.1

9 Electromagnetic Shielding Theory

Electromagnetic shielding is the primary defense against inductive coupling attacks. This section develops the theoretical framework for understanding shielding effectiveness and its limitations.

9.1 Shielding Effectiveness: Definition

The shielding effectiveness (SE) of an enclosure or barrier is defined as:

$$\mathrm{SE} = 20\log_{10}\left(\frac{E_{\text{unshielded}}}{E_{\text{shielded}}}\right) = 20\log_{10}\left(\frac{H_{\text{unshielded}}}{H_{\text{shielded}}}\right) \quad [\text{dB}] \tag{79}$$

For the attack scenario, the shielded EMF relates to the unshielded EMF as:

$$\mathcal{E}_{\text{shielded}} = \mathcal{E}_{\text{unshielded}} \times 10^{-\mathrm{SE}/20} \tag{80}$$

9.2 Shielding Mechanisms

Shielding effectiveness arises from three distinct physical mechanisms:

1. **Reflection Loss (R)**: Occurs at the interface between two media due to impedance mismatch. For a plane wave incident on a shield of impedance Z_s:

$$R = 20\log_{10}\left|\frac{(Z_w + Z_s)^2}{4Z_w Z_s}\right| \approx 168 + 10\log_{10}\left(\frac{\sigma_r}{\mu_r f}\right) \quad [\text{dB}] \tag{81}$$

   where $Z_w = \sqrt{\mu_0/\varepsilon_0} \approx 377\,\Omega$ is the wave impedance of free space.

2. **Absorption Loss (A)**: Depends on the skin depth and shield thickness t:

$$A = 20\log_{10}\left(e^{t/\delta}\right) = 8.69\frac{t}{\delta} = 8.69t\sqrt{\pi f \mu \sigma} \quad [\text{dB}] \tag{82}$$

3. **Multiple Reflection Loss (B)**: Significant only for thin shields ($t < \delta$):

$$B = 20\log_{10}\left(1 - e^{-2t/\delta}\right) \approx 20\log_{10}\left(\frac{2t}{\delta}\right) \quad [\text{dB}] \tag{83}$$

The total shielding effectiveness is:

$$\mathrm{SE}_{\text{total}} = R + A + B \tag{84}$$

9.3 Shielding Effectiveness of Common Materials

Table 4: Shielding effectiveness of common materials at 1 GHz for thickness $t = 0.5$ mm.

Material	σ (MS/m)	μ_r	δ (μm)	A (dB)	$\mathrm{SE}_{\text{total}}$ (dB)
Copper	58.0	1.0	2.09	2081	∼2100
Aluminum	37.7	1.0	2.59	1679	∼1700
Steel	1.0	100	1.59	2733	∼2800
Mu-metal	1.6	20000	0.08	54300	∼54500
Carbon fiber	0.01	1.0	50.3	86	∼100

Remark 9.1 (Practical Shielding Limitations)**.** *While the theoretical SE values in Table 4 appear extremely high, practical shielding effectiveness is limited by:*

1. ***Seams and joints**: Even a* 1 mm *gap can reduce SE by 20–40 dB*
2. ***Connectors**: Unshielded connectors are the weakest link*
3. ***Cable entry points**: Feedthroughs create leakage paths*
4. ***Low-frequency magnetic fields**: Mu-metal is required but expensive and brittle*
5. ***Ventilation openings**: Must be designed as waveguide-beyond-cutoff*

9.4 Skin Depth Analysis

The skin depth, defined in (34), determines the effective penetration depth of electromagnetic fields into a conductor. At frequencies much higher than the skin frequency $f_s = 1/(\pi\mu\sigma a^2)$, the current is confined to a thin layer near the surface.

For a copper conductor of radius $a = 0.5\,\text{mm}$:

$$f_s = \frac{1}{\pi \times 4\pi \times 10^{-7} \times 5.96 \times 10^7 \times (0.5 \times 10^{-3})^2} \approx 17\,\text{kHz} \tag{85}$$

This means that at all frequencies above 17 kHz (which includes all modern data communication frequencies), the current flows primarily on the surface of the conductor, which modifies the external field distribution compared to the DC case.

9.5 Shielding Effectiveness at Cable Interfaces

The most vulnerable point in a shielded cable system is the connector–cable interface. The transfer impedance Z_T of a shield characterizes the voltage induced per unit length per unit current flowing on the shield:

$$Z_T = R_{\text{DC}} \frac{\gamma t}{\sinh(\gamma t)} + j\omega M_{\text{braid}} \tag{86}$$

where R_{DC} is the DC resistance of the shield, $\gamma = (1+j)/\delta$ is the propagation constant in the shield material, and M_{braid} is the mutual inductance between braid apertures and the inner conductor.

10 Independent Component Analysis for Multi-Cable Separation

In realistic environments, the attacker's receiver captures a mixture of electromagnetic emissions from multiple cables and other sources. **Independent Component Analysis (ICA)** provides a principled mathematical framework for separating these mixed signals into their constituent sources.

10.1 The Linear Mixing Model

The observed signal vector $\boldsymbol{x}(t) \in \mathbb{R}^n$ is related to the source signal vector $\boldsymbol{s}(t) \in \mathbb{R}^n$ by:

$$\boldsymbol{x}(t) = \boldsymbol{A}\,\boldsymbol{s}(t) + \boldsymbol{n}(t) \tag{87}$$

where $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is an unknown mixing matrix and $\boldsymbol{n}(t)$ is additive noise. The goal of ICA is to find a demixing matrix $\boldsymbol{W}$ such that:

$$\hat{\boldsymbol{s}}(t) = \boldsymbol{W}\,\boldsymbol{x}(t) \approx \boldsymbol{s}(t) \tag{88}$$

10.2 Statistical Assumptions

ICA relies on the following fundamental assumptions:

1. **Statistical Independence**: The source signals $s_i(t)$ are statistically mutually independent.
2. **Non-Gaussianity**: At most one source signal can be Gaussian (by the Central Limit Theorem).
3. **Square Mixing**: The number of sensors equals the number of sources ($n = m$).
4. **Non-singular Mixing**: The mixing matrix $\boldsymbol{A}$ must be invertible.

Remark 10.1 (Applicability to EM Side-Channel Attacks)**.** *Digital signals are inherently non-Gaussian (they are discrete-valued), and emissions from different cables carrying independent data streams are statistically independent. These conditions are well-satisfied in the EM side-channel attack scenario, making ICA a powerful tool for multi-cable environments.*

10.3 Preprocessing: Whitening

Before applying ICA, the observed signals are whitened (sphered) using the eigenvalue decomposition of the covariance matrix:

$$\boldsymbol{C}_x = \mathbb{E}[\boldsymbol{x}\boldsymbol{x}^T] = \boldsymbol{V}\,\boldsymbol{\Lambda}\,\boldsymbol{V}^T \tag{89}$$

The whitening transformation is:

$$\boldsymbol{z} = \boldsymbol{D}\,\boldsymbol{x}, \qquad \boldsymbol{D} = \boldsymbol{\Lambda}^{-1/2}\,\boldsymbol{V}^T \tag{90}$$

so that $\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^T] = \boldsymbol{I}$.

10.4 FastICA Algorithm

The FastICA algorithm maximizes the non-Gaussianity of the projected components using negentropy as the objective function:

$$J(y) = H(y_{\text{gauss}}) - H(y) \tag{91}$$

where $H(y) = -\int p(y)\log p(y)\,\mathrm{d}y$ is the differential entropy. Using a fixed-point iteration for each component:

$$\boldsymbol{w}^+ = \mathbb{E}[\boldsymbol{z}g(\boldsymbol{w}^T\boldsymbol{z})] - \mathbb{E}[g'(\boldsymbol{w}^T\boldsymbol{z})]\boldsymbol{w} \tag{92}$$

$$\boldsymbol{w} = \frac{\boldsymbol{w}^+}{\|\boldsymbol{w}^+\|} \tag{93}$$

where g is a nonquadratic contrast function. Common choices include:

- $g(u) = \tanh(u)$: suitable for super-Gaussian sources
- $g(u) = u\exp(-u^2/2)$: suitable for sub-Gaussian sources
- $g(u) = u^3$: kurtosis-based (sensitive to outliers)

11 Transmission Line Theory for Cable Radiation

Understanding how currents in cables generate radiating electromagnetic fields requires a rigorous treatment of transmission line theory extended to include radiation effects.

11.1 Current Mode Analysis

Currents on cable systems can be decomposed into two fundamental modes:

11.1.1 Differential Mode (DM)

The differential mode current flows in opposite directions on the signal and return conductors:

$$I_{\text{DM}} = \frac{I_{\text{signal}} - I_{\text{return}}}{2} \tag{94}$$

Differential mode currents generate **electric dipole-type** radiation with the field proportional to:

$$|E_{\text{DM}}| \propto f \cdot l \cdot |I_{\text{DM}}| \cdot \frac{1}{r} \tag{95}$$

11.1.2 Common Mode (CM)

The common mode current flows in the **same direction** on all conductors and returns through the ground path:

$$I_{\text{CM}} = \frac{I_{\text{signal}} + I_{\text{return}}}{2} \tag{96}$$

Common mode currents generate much stronger radiation because they act as a monopole antenna:

$$|E_{\text{CM}}| \propto f \cdot l \cdot |I_{\text{CM}}| \cdot \frac{1}{r} \tag{97}$$

Proposition 11.1 (Common Mode Dominance in Side-Channel Attacks)**.** *Even a small common mode current ($I_{CM} \ll I_{DM}$) can produce radiation comparable to or exceeding the differential mode radiation. Typically, $I_{CM}/I_{DM} \approx -20$ to $-60\,dB$ (i.e., 0.1% to 0.001%), but the common mode radiation efficiency is 10–100 times higher. In the side-channel attack context, common mode emissions are often the dominant component of the captured signal.*

11.2 Antenna Model for Cable Emissions

A cable of length l carrying common mode current I_{CM} can be modeled as a linear antenna. The radiated electric field at distance r in the far field ($r \gg \lambda$) is:

$$E_\theta = \frac{j\eta_0 I_{\text{CM}}}{2\pi r} e^{-jkr} \frac{\cos(kl\cos\theta/2) - \cos(kl/2)}{\sin\theta} \tag{98}$$

where $\eta_0 = \sqrt{\mu_0/\varepsilon_0} \approx 377\,\Omega$ is the free-space impedance and $k = 2\pi/\lambda$ is the wavenumber.

For electrically short cables ($kl \ll 1$), this simplifies to the Hertzian dipole formula:

$$E_\theta \approx \frac{j\eta_0 k I_{\text{CM}} l \sin\theta}{4\pi r} \tag{99}$$

Theorem 11.2 (Maximum Radiated Power)**.** *The total power radiated by a Hertzian dipole is:*

$$P_{rad} = \frac{\eta_0 (kl)^2 |I_{CM}|^2}{12\pi} = \frac{\eta_0 \pi}{3} \left(\frac{l}{\lambda}\right)^2 |I_{CM}|^2 \tag{100}$$

For $l = 1\,\mathrm{m}$*,* $f = 500\,\mathrm{MHz}$ *(*$\lambda = 0.6\,\mathrm{m}$*), and* $I_{CM} = 1\,\mu\mathrm{A}$*:*

$$P_{rad} = \frac{377\pi}{3} \left(\frac{1}{0.6}\right)^2 (10^{-6})^2 \approx 1.1\,\mathrm{pW} \tag{101}$$

This tiny radiated power is nevertheless detectable with sensitive receivers.

12 Biot–Savart Law and Near-Field Analysis

The near-field regime is particularly relevant for electromagnetic side-channel attacks, as the attacker is typically positioned within a few wavelengths of the victim cable.

12.1 Biot–Savart Law

The magnetic field at position $\boldsymbol{r}$ due to a current distribution $\boldsymbol{J}(\boldsymbol{r}')$ is:

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{r}') \times (\boldsymbol{r} - \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|^3} \mathrm{d}V' \tag{102}$$

For a thin wire carrying current I along path C:

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0 I}{4\pi} \oint_C \frac{\mathrm{d}\boldsymbol{\ell}' \times (\boldsymbol{r} - \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|^3} \tag{103}$$

12.2 Magnetic Field of a Finite Straight Wire

For a straight wire segment of length L carrying current I, the magnetic field at a perpendicular distance d from the wire is:

$$B = \frac{\mu_0 I}{4\pi d} (\sin\alpha_1 + \sin\alpha_2) \tag{104}$$

where α_1 and α_2 are the angles subtended by the wire endpoints as seen from the observation point. For an infinite wire ($\alpha_1 = \alpha_2 = \pi/2$), this reduces to the standard result (18).

12.3 Near-Field vs. Far-Field Transitions

The electromagnetic field around a radiating structure can be divided into three regions:

1. **Reactive Near Field** ($r < \lambda/2\pi$): Energy is predominantly stored, not radiated. The electric and magnetic fields are not in phase. The impedance is either very high (electric dipole) or very low (magnetic dipole).

2. **Radiating Near Field (Fresnel Region)** ($\lambda/2\pi < r < 2D^2/\lambda$): Radiation begins to dominate. The field pattern depends on distance.

3. **Far Field (Fraunhofer Region)** ($r > 2D^2/\lambda$): The field pattern is independent of distance. Electric and magnetic fields are in phase and related by $\boldsymbol{E} = \eta_0 \boldsymbol{H} \times \hat{\boldsymbol{r}}$.

Remark 12.1 (Attack Implications)**.** *For side-channel attacks on cables operating at frequencies up to* $1\,\mathrm{GHz}$ *(*$\lambda = 30\,\mathrm{cm}$*), the reactive near field extends to approximately* $d \approx 5\,\mathrm{cm}$*. Since typical attack distances range from a few centimeters to several meters, the attacker may operate in any of these regimes, requiring different models for signal propagation. In the reactive near field, the magnetic field falls off as* $1/r^2$ *to* $1/r^3$*, while in the far field it falls off as* $1/r$*.*

13 Poynting Vector and Energy Flow

13.1 The Poynting Theorem

The Poynting vector describes the directional energy flux density of the electromagnetic field:

$$\boldsymbol{S} = \boldsymbol{E} \times \boldsymbol{H} \quad [\mathrm{W\,m^{-2}}] \tag{105}$$

The Poynting theorem states:

$$-\oint_S \boldsymbol{S} \cdot \mathrm{d}\boldsymbol{A} = \int_V \left(\boldsymbol{J} \cdot \boldsymbol{E} + \frac{\partial}{\partial t} \left(\frac{1}{2}\varepsilon E^2 + \frac{1}{2}\mu H^2 \right) \right) \mathrm{d}V \tag{106}$$

The left side represents the net power flowing out of the volume V, while the right side represents the ohmic power dissipation plus the rate of change of stored electromagnetic energy.

13.2 Time-Averaged Power Flow

For sinusoidal signals with complex phasors $\boldsymbol{E}$ and $\boldsymbol{H}$, the time-averaged Poynting vector is:

$$\langle \boldsymbol{S} \rangle = \frac{1}{2} \mathrm{Re}(\boldsymbol{E} \times \boldsymbol{H}^*) \tag{107}$$

For the induced signal in the attack scenario, the power captured by the receiving loop of area A_{rx} at distance d from the victim cable is:

$$P_{\mathrm{rx}} = \langle \boldsymbol{S} \rangle \cdot A_{\mathrm{rx}} = \frac{|\mathcal{E}|^2}{4R_{\mathrm{rx}}} \tag{108}$$

where R_{rx} is the input impedance of the receiver.

14 Cryptographic Implications of Physical Leakage

The electromagnetic side-channel attack has profound implications for cryptographic security. This section analyzes the theoretical connection between physical signal leakage and key extraction.

14.1 Information-Theoretic Framework

The mutual information between the victim's data X and the attacker's observation Y quantifies the maximum extractable information:

$$I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) \tag{109}$$

where $H(\cdot)$ denotes Shannon entropy and $H(\cdot|\cdot)$ denotes conditional entropy. For an n-bit key:

$$H(X) = n \text{ bits (for a uniform random key)} \tag{110}$$

The residual uncertainty after observation is $H(X|Y)$, and the information leakage is $I(X;Y)$. A complete key recovery requires $I(X;Y) = n$ bits.

14.2 Side-Channel Attacks on Cryptographic Implementations

14.2.1 Simple Power Analysis (SPA)

SPA exploits direct visualization of power consumption patterns during cryptographic operations. The electromagnetic analogue involves observing the EMF envelope during key-dependent operations:

$$\mathcal{E}_{\text{SPA}}(t) = \sum_{i=1}^{N_{\text{ops}}} h_i(t - t_i) \cdot f(k_i) \tag{111}$$

where h_i is the EMF signature of operation i, t_i is its timing, and $f(k_i)$ is a key-dependent function.

14.2.2 Differential Power Analysis (DPA)

DPA uses statistical techniques to extract small key-dependent differences from noisy measurements:

$$\Delta_D = \mathbb{E}[\mathcal{E}(t)|D = 0] - \mathbb{E}[\mathcal{E}(t)|D = 1] \tag{112}$$

where $D = g(m, k)$ is a selection function depending on known plaintext m and guessed key bit k. The correct key hypothesis produces a statistically significant Δ_D, while incorrect hypotheses yield $\Delta_D \approx 0$.

Under the null hypothesis (incorrect key guess), the test statistic follows:

$$\Delta_D \sim \mathcal{N}\left(0, \frac{2\sigma^2}{N/2}\right) = \mathcal{N}\left(0, \frac{4\sigma^2}{N}\right) \tag{113}$$

where N is the total number of traces and σ^2 is the noise variance.

Proposition 14.1 (Number of Traces for DPA)**.** *For a detection threshold α (significance level), the minimum number of traces required is:*

$$N_{\min} = \frac{4(z_{\alpha/2})^2\sigma^2}{\Delta^2} \tag{114}$$

where Δ is the expected signal difference between the correct and incorrect key hypotheses and $z_{\alpha/2}$ is the standard normal quantile.

15 Conclusion and Open Research Directions

This document has presented a comprehensive theoretical treatment of the electromagnetic side-channel attack through inductive coupling, establishing the complete chain of physical and mathematical reasoning from Maxwell's equations to practical attack feasibility. The key theoretical results can be summarized as follows.

15.1 Summary of Key Theoretical Results

R1: The electromagnetic side-channel attack is fundamentally governed by **Faraday's law of induction** ((16)), which establishes that the induced EMF in the receiver is proportional to the **time derivative** of the victim current.

R2: The **mutual inductance** between parallel cables ((31)) determines the coupling strength and depends logarithmically on the geometric parameters (cable length, separation distance).

R3: The induced signal encodes **bit transitions** rather than absolute bit values ((40)), requiring integration-based reconstruction algorithms.

R4: In the frequency domain, the induced signal power scales as f^2 ((62)), making high-speed data links inherently more vulnerable.

R5: The attack feasibility is governed by the **signal-to-noise ratio** ((73)), with a minimum SNR of approximately 10 dB required for practical key recovery.

R6: Electromagnetic shielding ((84)) provides effective but imperfect protection, with practical limitations arising from seams, connectors, and cable entry points.

15.2 Open Research Problems

Several fundamental research problems remain open in this domain:

P1: **Long-range detection**: The $1/r$ decay of far-field radiation and $1/r^3$ decay of near-field magnetic coupling impose fundamental limits on attack range. Developing signal processing techniques (e.g., coherent averaging, distributed sensor arrays) to extend the effective range beyond 10 m remains an open challenge.

P2: **Multi-cable separation in reverberant environments**: Current ICA-based approaches (Section 10) assume linear mixing in free space. Extending these techniques to handle reflections, multipath propagation, and non-linear mixing in realistic indoor environments is an active area of research.

P3: **Encrypted traffic analysis**: While the electromagnetic signature of encrypted traffic leaks metadata (packet timing, size distribution), developing theoretical bounds on the information extractable from these side channels requires further work in information theory and cryptography.

P4: **Unified countermeasure verification**: No standardized framework exists for quantitatively evaluating the electromagnetic security of cable systems. Developing such a framework, analogous to the Common Criteria for software security, would be a significant contribution.

P5: Quantum effects: At extremely short ranges and high frequencies, quantum mechanical effects may modify the electromagnetic coupling. The implications of quantum electrodynamics (QED) for side-channel attacks remain unexplored.

References

[1] J. C. Maxwell, "A Dynamical Theory of the Electromagnetic Field," *Philosophical Transactions of the Royal Society of London*, vol. 155, pp. 459–512, 1865.

[2] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. New York: Wiley, 1999.

[3] D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed. Cambridge: Cambridge University Press, 2017.

[4] W. van Eck, "Electromagnetic Radiation from Video Display Units: An Eavesdropping Risk?" *Computers & Security*, vol. 4, no. 4, pp. 269–286, 1985.

[5] P. C. Kocher, J. Jaffe, and B. Jun, "Differential Power Analysis," in *Advances in Cryptology — CRYPTO '99*, Springer, 1999, pp. 388–397.

[6] M. Guri, Y. Solewicz, A. Daidakulov, and Y. Elovici, "USBee: Air-Gap Covert-Channel via Electromagnetic Emission from USB," in *Proc. 16th Annual Conference on Privacy, Security and Trust (PST)*, 2018.

[7] M. Guri, D. Zadok, A. Daidakulov, and Y. Elovici, "ODINI: Bridging the Air-Gap between Isolated Networks and Mobile Phones using Magnetic Fields," *arXiv preprint arXiv:1808.03468*, 2018.

[8] J. Loughry and D. A. Umphress, "Information Leakage from Optical Emanations," *ACM Transactions on Information and System Security*, vol. 5, no. 3, pp. 262–289, 2002.

[9] F. W. Grover, *Inductance Calculations: Working Formulas and Tables*. Mineola, NY: Dover Publications, 2004.

[10] C. R. Paul, *Introduction to Electromagnetic Compatibility*, 2nd ed. Hoboken, NJ: Wiley, 2006.

[11] F. M. Tesche, M. V. Ianoz, and T. Karlsson, *EMC Analysis Methods and Computational Models*. New York: Wiley, 1992.

[12] A. Hyvärinen, J. Karhunen, and E. Oja, *Independent Component Analysis*. New York: Wiley, 2001.

[13] D. R. Stinson, *Cryptography: Theory and Practice*, 3rd ed. Boca Raton, FL: CRC Press, 2006.

[14] M. Vuagnoux and S. Pasini, "Compromising Electromagnetic Emanations of Wired and Wireless Keyboards," in *Proc. 18th USENIX Security Symposium*, 2009, pp. 1–16.

[15] R. Cheggou, "From Geometrical Optics to Wave Optics: Maxwell's Theory," ENSTA Algiers MI2 Course Materials, 2025.

[16] Various Authors, "Deep Learning for Electromagnetic Side-Channel Analysis," *IEEE Transactions on Information Forensics and Security*, vol. 18, 2023.